

Exercise 1. Determine whether the following function is a probability density function:

$$f(x) = \begin{cases} 0.25, & -2 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Exercise 2. Find a constant k such that the function becomes a probability density function:

$$f(x) = \begin{cases} k(2x + 1), & 0 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

Exercise 3. Determine whether the following function is a PDF:

$$f(x) = \begin{cases} 1 - x, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

Exercise 4. Let

$$f(x) = \begin{cases} kx^3, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Find k so that f is a PDF.
- (b) Compute $\mathbb{P}(1 \leq X \leq 2)$.

Exercise 5. A random variable X has density

$$f(x) = \begin{cases} \frac{1}{6}(3 - x), & 0 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Verify that f is a PDF.
- (b) Find $\mathbb{P}(X > 1)$.

Exercise 6. The waiting time (in hours) for customer support follows an exponential distribution with parameter $\lambda = 0.5$:

$$f(t) = \begin{cases} 0.5e^{-0.5t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

- (a) Find $\mathbb{P}(T \leq 4)$.
- (b) Find $\mathbb{P}(T > 6)$.

Exercise 7. A delivery time X is uniformly distributed on $[5, 25]$ minutes.

- (a) Write the density function.
- (b) Find $\mathbb{P}(X < 10)$.
- (c) Find $\mathbb{P}(15 \leq X \leq 20)$.

Exercise 8. Let

$$f(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Verify that f is a PDF.
- (b) Compute the expected value $\mathbb{E}(X)$.

Exercise 9. The random variable X has density

$$f(x) = \begin{cases} \frac{3}{4}(1 - x^2), & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Verify that f is a PDF.
- (b) Find $\mathbb{E}(X)$.

Exercise 10. The height of plants in a greenhouse is normally distributed with mean $\mu = 150$ cm and standard deviation $\sigma = 12$ cm.

- (a) Write the normal density function.
- (b) Set up the integral for $\mathbb{P}(138 \leq X \leq 162)$.
- (c) Estimate the probability using the empirical rule.